

The Joint Production of Price Efficiency and Heavy-Tailed Volatility in Minimal Agent-Based Markets: Evidence from Three Pre-Registered Refutations

Abstract

Agent-based models of financial markets typically reproduce the stylized facts of asset returns piecemeal: models are exhibited that match subsets of the facts, and failed attempts on the remainder are rarely reported. This paper scores minimal flow-based markets against a fixed ten-event battery calibrated so that real equity data passes all ten events and geometric Brownian motion exactly three. A market of fundamentalists, chartists, volatility-targeting investors, and market makers attains five of ten events, and its failures are diagnostic: the missing events concern long memory, time-irreversibility, and the information content of price signs. Three pre-registered experiments with pre-specified refutation criteria then test whether added rationality can close the gap: (i) an anticipatory agent that front-runs the forecastable flow; (ii) the fixed point of mutual anticipation, iterated to depth five; (iii) a price-setting market maker that provably absorbs the forecastable component into the price level. All three attempts are refuted. Deeper anticipation increases the residual sign predictability of returns (median 0.0184 to 0.0257 bits) while contracting the flow it targets, and exact absorption removes the heavy tails (kurtosis $8.8 \rightarrow 3.0$), the leverage effect ($-0.125 \rightarrow -0.008$), and volatility clustering together with the predictability. We conclude that, in this model class, sign efficiency and the heavy-tailed, asymmetric volatility regularities are jointly produced by the same state-dependent flows and cannot be obtained separately. Cross-venue evidence — the leverage effect ordered by market structure across equities (-0.04), foreign exchange (≈ 0), and cryptocurrencies ($+0.03$) — is consistent with the proposed mechanism. All results are pre-registered, validated against synthetic ground truth, and regenerable from the repository.

Keywords: stylized facts; agent-based models; market efficiency; price impact; leverage effect; pre-registered research.

JEL classification: G14; G17; C63; C52.

1 Introduction

Daily asset returns exhibit two properties simultaneously. Their signs are nearly unforecastable from public information, and their amplitudes display the heavy-tailed, asymmetric regularities documented across markets and decades: excess kurtosis, volatility clustering, the leverage effect, and gain/loss tail asymmetry (Cont, 2001). Throughout the paper we refer to the first property as *sign efficiency* and to the second group — the amplitude-side stylized facts — as the *amplitude facts*. This paper asks whether a minimal market mechanism can generate both properties at once, and answers by refutation: in flow-based minimal markets, sign efficiency and the amplitude facts are jointly produced by the same flows, and three pre-registered attempts

to obtain them separately — rationality added as a trading agent, as the fixed point of mutual anticipation, and as a price-setting rule — all fail against refutation criteria declared in advance. The refutations are the result: they compose into a structural statement about what this model class cannot do, measured on a fixed instrument.

Three research questions organize the analysis. First, which of the stylized facts can a minimal flow-based market reproduce when all of them are scored jointly on a fixed, pre-calibrated instrument rather than exhibited selectively? Second, can bounded rationality — introduced as a trading agent, iterated toward its fixed point, or embedded in the pricing rule itself — supply the facts such a market misses, in particular the near-absence of sign information in prices? Third, is the mechanism that answers the first two questions consistent with how the amplitude facts vary across real trading venues with different flow structures?

The questions sit in a literature with a shared target list: the stylized facts that any candidate mechanism for price formation should reproduce (Cont, 2001). Agent-based modeling took up the challenge early: heterogeneous-agent markets with fundamentalists and chartists generate fat tails and clustered volatility from agent interaction rather than from exogenous news (Lux and Marchesi, 1999), and a substantial literature has since mapped which ingredients produce which facts (LeBaron, 2006). Two gaps in that literature motivate the design here. First, scoring is usually piecemeal: a model is exhibited that matches two or three facts, with the remaining facts unexamined, so the literature accumulates existence proofs rather than a ledger of what a given mechanism class can and cannot reproduce. Second, negative results are structurally underproduced: when an added ingredient fails to improve a model, the failure is rarely reported, almost never pre-registered, and therefore carries no evidential weight. We address both gaps with a fixed measurement instrument and a pre-registered refutation chain.

The paper makes four contributions. *First*, it introduces a ten-event battery of binary stylized-fact tests, computed identically on real and simulated series, calibrated at both ends of its range: real S&P 500 index data (SPY) passes 10/10 events while geometric Brownian motion passes exactly 3/10, so intermediate scores are interpretable (Section 4.1). *Second*, it establishes, through three successive pre-registered experiments with pre-specified refutation criteria and parameter budgets, that the best flow-only configuration’s score of 5/10 is invariant to added rationality of three distinct kinds — a trading anticipator, the iterated fixed point of mutual anticipation, and a price-setting market maker — and that the refutations compose into a structural claim: in this model class the residual sign information and the amplitude facts are generated by the same state-dependent flows (Section 6). *Third*, it develops a methodology that makes such negative results evidential: every estimator and every tested mechanism must pass a size and power validation against synthetic data-generating processes with known properties before any evaluation data is read, so a refutation is a measurement rather than an absence of result (Section 4). *Fourth*, it reports a cross-venue measurement of the leverage effect with a single estimator battery — equities -0.04 , foreign exchange statistically zero, cryptocurrencies $+0.03$ in point estimate — whose ordering is consistent with the flow-structure mechanism, together with an explicit power analysis of what the cross-venue data can and cannot yet certify (Section 7).

The headline mechanism result can be stated compactly. Adding one layer of anticipation to the minimal market leaves its battery score unchanged and moves the sign leak — the residual forecastability of next-day return signs, defined precisely in Section 6 — one derivative earlier; iterating anticipation five layers deep contracts the model-space forecastable flow by a factor of four while the leaked sign information in realized prices moves the opposite way (median

direction bits $0.0184 \rightarrow 0.0200 \rightarrow 0.0257$; the depth-five rise holds on 7 of 8 paired seeds); and a maker that removes the forecastable component’s price impact exactly — verified to the identity on a synthetic world — does not close the leak either, but instead deletes the fat tails (kurtosis $8.8 \rightarrow 3.0$), the leverage effect ($-0.125 \rightarrow -0.008$), and volatility clustering together with the amplitude structure the flows were generating. Sign efficiency and the amplitude facts are not separate dials in this model class.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the measurement methodology: the event battery, the size/power validation protocol, and the pre-registration design. Section 5 specifies the base market model and the three rationality mechanisms. Section 6 reports the results of the ablation study and the three experiments. Section 7 presents the cross-venue evidence. Section 8 states the scope of the claims and the limitations, and Section 9 concludes. Appendices A–C contain exact threshold definitions, the multi-venue calibration audit, and robustness material.

2 Related work

2.1 Stylized facts of asset returns

Heavy tails in speculative price changes were documented by Mandelbrot (1963); Cont (2001) fixed the canonical list of stylized facts — heavy tails, volatility clustering, the leverage effect, near-absence of linear return autocorrelation, gain/loss asymmetry — that this paper’s battery operationalizes, and our contribution to that program is calibration at both ends (real data 10/10, GBM exactly 3/10) so that partial scores are interpretable. In reduced form, volatility clustering is captured by the ARCH family (Engle, 1982; Bollerslev, 1986), with asymmetric extensions such as GJR-GARCH (Glosten et al., 1993) adding the leverage channel. The battery deliberately includes an event this family fails: amplitude correlations in real data decay hyperbolically rather than exponentially (Ding et al., 1993), a message sharpened by the rough-volatility literature (Gatheral et al., 2018). The measurement program behind this paper replicated a Hurst exponent $H \approx 0.10$ on 16 years of SPY range-based volatility, and event E4 is calibrated so that exponential-memory processes (GJR-GARCH) fail it — which is exactly the event no flow configuration and no rationality layer ever produced, consistent with rough volatility’s message that the clock’s memory is structural, not flow-generated. Time-reversal asymmetry, the target of event E8, is a documented stylized fact in its own right (Zumbach, 2009). Outside equities, the stylized-facts program has been extended to cryptocurrency markets (Urquhart, 2016; Zhang et al., 2018), whose distinct flow structure Section 7 exploits.

2.2 Heterogeneous-agent models

The agent-based literature descending from Lux and Marchesi (1999) and surveyed by LeBaron (2000, 2006) and Hommes (2006) established that fundamentalist–chartist interaction generates fat tails and clustering, whether through switching between predictor types, herding in the communication structure (Cont and Bouchaud, 2000), or the destabilizing feedback of technical strategies on the price they condition on (Farmer and Joshi, 2002; Chiarella et al., 2009). Our ablation reproduces the classical result (and its fragility: chartists without market makers destroy efficiency, scoring 1/10) but differs in asking the complement — which facts a flow-based

market *cannot* reproduce at any rationality depth, established by pre-registered refutation rather than by exhibition.

Two adjacent traditions ask closely related questions with different instruments. In the heterogeneous-beliefs models of Brock and Hommes (1997, 1998), the sophistication of agents' expectations is a bifurcation parameter of a deterministic pricing map: making agents more forward-looking changes the dynamics qualitatively rather than monotonically improving them. Our K -ladder is a stochastic, capital-bounded analogue of that exercise, and it finds the same non-monotonicity in a measured quantity — the sign leak grows as anticipation deepens. The minority-game literature (Challet and Zhang, 1997; Challet et al., 2004) and the intermittent agent-based markets of Giardina and Bouchaud (2003) derive stylized facts from the statistical mechanics of adaptive agents competing over a scarce resource, with realistic regimes appearing near phase boundaries; our battery differs in scoring a fixed list of calibrated events rather than exhibiting a regime, but the joint-production finding is consistent with that literature's lesson that efficiency and excess volatility are properties of one collective state, not separate features.

2.3 Market efficiency and the sign channel

The battery's E9 event operationalizes the oldest target in the field. Samuelson (1965) proved that properly anticipated prices fluctuate randomly, and Fama (1970) organized the empirical program around informational efficiency. A classical argument holds that when information is costly, full informational efficiency cannot be an equilibrium: some residual predictability must survive to compensate the agents who remove it. The joint-production result is a mechanical relative of that argument with the cost relocated: in this model class the residual sign information is not a fee paid to arbitrageurs but a by-product of the same state-dependent flows that generate the volatility structure, and the three experiments show it cannot be traded, iterated, or quoted away without taking the amplitude facts with it.

2.4 The leverage effect

The classical explanations of the equity leverage effect are financial leverage in the firm's capital structure (Christie, 1982) and volatility feedback through the equity risk premium (Bekaert and Wu, 2000); Bouchaud et al. (2001) measured the effect's retarded temporal structure across individual stocks and indices and modeled it directly at the price-process level. Both classical mechanisms are equity-specific, which is what makes the cross-venue comparison of Section 7 informative: a currency is not a levered claim on productive assets and carries no equity risk premium, so whatever return–volatility asymmetry survives outside equities must have another source.

2.5 Order flow, price impact, and de-leveraging feedback

The price-impact literature of Bouchaud and coauthors (Bouchaud et al., 2009, 2018) underlies the paper's central object: the idea that order flow is the proximate mover of prices and that liquidity provision digests predictable flow. Our quote-skewing maker is a minimal formalization of perfect digestion — and Experiment III's finding that exact absorption deletes the volatility dynamics along with the leak is, in that vocabulary, a statement that the market's amplitude structure *is* its undigested flow. The de-leveraging spiral at the model's core is a minimal form of the funding-liquidity feedback of Brunnermeier and Pedersen (2009) and of the procyclical

intermediary leverage documented by Adrian and Shin (2010): a volatility rise forces balance-sheet contraction, and the contraction moves prices. Volatility-targeting mandates, which hard-wire this feedback into a mechanical trading rule, are the model’s concrete instance of the mechanism.

2.6 Reflexivity and endogeneity

Filimonov and Sornette (2012) measured near-critical reflexivity in markets via the Hawkes branching ratio, and Hardiman et al. (2013) showed the estimate is sensitive to kernel specification, finding near-critical values throughout two decades of data. The measurement program behind this paper found the raw result replicates (branching ratio 0.68 across 48 US assets) but collapses to 0.25 — a no-self-excitation null — once events are deformed by the volatility clock, so most apparent reflexivity at the daily horizon is volatility clustering; the same deformation-first discipline shapes the battery’s E8 event.

2.7 Multiple testing and pre-registration in finance

Harvey et al. (2016) documented the multiple-testing problem behind the factor literature and argued that published findings should clear substantially higher evidential hurdles; Bailey and López de Prado (2014) and Bailey et al. (2017) supplied the deflated Sharpe ratio and the probability of backtest overfitting as concrete corrections. The protocol of Section 4.3 — pre-registered hypotheses, bounded search budgets, deterministic tie-breaks, and refutation criteria honored when they fire — is that program’s counterpart for mechanism research rather than performance research: it is what makes the negative results of Section 6 reportable as measurements.

3 Data and descriptive statistics

All series are daily. Four data sets enter the paper: the US equity calibration series, the multi-index audit set, the foreign-exchange panel, and the cryptocurrency panel. All were obtained from Yahoo Finance as daily open/high/low/close (OHLC) bars and are cached in the repository’s data layer; every screen described below runs inside the loaders, so a failing screen aborts the study rather than silently degrading it.

3.1 Equities

The battery’s calibration anchor is SPY, the S&P 500 index ETF, over 2010–2026 (close-only, approximately 4,100 trading days). The event thresholds themselves descend from a prior measurement program on ~48 liquid US equities and ETFs over the same span (Section 4.1); panels in that program require 95% coverage per instrument and forward-fill gaps of at most three days. The multi-index audit set (Appendix B) comprises six further index ETFs on a fixed, registered list: QQQ, DIA, and IWM (US, 2010–2026) and three locally-listed non-US index trackers — 1306.T (TOPIX), EXW1.DE (EURO STOXX 50), and 2800.HK (Hang Seng). The cross-venue equity cohort of Section 7 consists of four universes of locally-listed large-capitalization equities (US, Japan, Europe, Asia-EM), each in a single time-zone block, from the research program’s transfer study.

Panel	N	Span	kurtosis	kurtosis (deformed)	leverage effect
US equities	49	2010–2026	13.1	3.4	−0.047
Japan equities	29	2010–2026	8.9	3.3	−0.039
Europe equities	36	2010–2026	11.3	3.6	−0.046
Asia-EM equities	29	2010–2026	8.4	3.8	−0.030
FX crosses	13	2010–2026	7.6	2.7	+0.005
Cryptocurrencies	10	2017–2026	16.6	4.5	+0.031

Table 1. Universe-level descriptive statistics. N is the number of instruments on each registered list (per-instrument coverage screens may drop instruments from individual analyses). Statistics are the measurement program’s pooled universe-level estimates: kurtosis of daily returns, kurtosis of volatility-deformed returns $r_t/\hat{\sigma}_t$, and the leverage effect $\text{corr}(r_t, \text{RV}_{t+1..10})$. The deformed-kurtosis column collapses toward the Gaussian in every venue; the leverage column changes sign across venues.

3.2 Foreign exchange

The FX panel contains 13 liquid crosses — twelve majors plus USDMXN as a pre-declared emerging-market stress case — over 2010–2026, the same span as the equity universes. Yahoo FX bars occasionally carry degenerate ranges (high equal to low, or the range pinned to $|\text{close} - \text{open}|$), which would invalidate range-based variance estimation (Garman and Klass, 1980); a screen measured the fraction of days with a real intraday range per pair before the study design was locked. All 13 pairs pass the pre-declared 95% bar (measured real-range fractions 99.88–100.00%), and none were dropped.

3.3 Cryptocurrencies

The registered cryptocurrency panel contains the 10 largest coins with full-period daily history (BTC, ETH, XRP, LTC, BCH, ADA, DOGE, LINK, XLM, ETC) over 2017-11-09 to 2026-06-04, the span for which major-coin daily data exist; every coin passes the same real-range screen at 100% of days. A registered robustness widening to 17 majors (adding TRX, XMR, EOS, NEO, DASH, ZEC, BAT; ATOM excluded by a pre-declared history rule) is used in the power analysis of Section 7 and leaves the registered 10-coin panel untouched.

3.4 Descriptive statistics

Table 1 reports universe-level descriptive statistics for the six panels used in the cross-venue analysis. Two regularities in the table organize the paper. First, the kurtosis of volatility-deformed returns sits near the Gaussian value in every venue while raw kurtosis does not: conditionally on their own volatility path, daily returns are close to Gaussian everywhere, which is what the battery’s event E6 tests in simulation. Second, the leverage effect changes sign across venues — negative in every equity universe, indistinguishable from zero in FX, positive in point estimate in cryptocurrencies — and that ordering is the external-validity evidence analyzed in Section 7. The full statistic vector for the battery’s calibration anchor (SPY) appears in Table 9 of Appendix A.

#	Event	Pass criterion
E1	price efficiency	$AC_1(r) \in [-0.15, 0.05]$ (allows the real index’s mild daily reversal)
E2	fat tails	$kurtosis(r) \in [4.5, 40]$
E3	vol clustering	$AC_1(r) \geq 0.12$ and $\text{mean } AC(r , \text{lags } 5\text{--}20) \geq 0.05$
E4	long-memory clock	8-week volatility-clock level autocorrelation above threshold
E5	leverage effect	$\text{mean } \text{corr}(r_t, RV_{t+1..10}) \leq -0.03$
E6	one-clock Gaussianization	$kurtosis(r/\sigma_{5d}) \leq 5$
E7	clock jumps	AR(1) clock-innovation skew beyond the $\log\text{-}\chi^2$ noise floor
E8	arrow of time in the coupling	entropy production significant in r and reduced $\geq 25\%$ by vol deformation
E9	no sign information	direction bits $\leq$ shuffle null (95%)
E10	tail asymmetry	gain/loss tail asymmetry at 2.5σ

Table 2. The ten-event battery (final calibrated form; exact estimator definitions and thresholds in Appendix A). E1, E4, E7 and E10 were recalibrated against real data before any ablation was run, under a gate requiring real SPY to score 10/10 and geometric Brownian motion exactly 3/10; notably, a GJR-GARCH process (Glosten et al., 1993) fails E4 — exponential memory cannot mimic long memory. E9 measures the mutual information between three public tape features and the next-day return sign with a discrete plug-in estimator against a permutation null.

4 Methodology

4.1 The ten-event battery

The battery consists of ten binary events, computed identically on real data and on simulated output, with pass criteria fixed before any agent-based run. Table 2 lists them; Appendix A gives the exact statistic definitions and thresholds as implemented, together with the full statistic vector at both calibration ends. The events are close-only versions of regularities measured and gate-validated in a prior program on ~ 48 liquid US equities and ETFs (2010–2026), so the thresholds are empirical ranges, not conventions.

Calibration pins both ends of the scale: real SPY (close-only) scores 10/10; a pure GBM scores exactly 3/10, passing only E1, E6 and E9 — Gaussian worlds are efficient, trivially one-clock, and carry no sign information. Everything between 3 and 10 is what a candidate mechanism must supply. Two scoping remarks apply. The thresholds are empirical ranges from a US program, and a registered audit on six further indices (Table 10 in Appendix B) measures how far the joint anchor extends: DIA also scores 10/10, QQQ and IWM 8/10, while three non-US index ETFs score only 5–7/10 — the battery is a US-calibrated instrument, and its function in this paper is comparative — every configuration is scored on the same events with the same seeds — not to certify absolute realism for any venue. Simulated scores carry a seed-bootstrap standard error — the eight evaluation seeds are resampled with replacement 2000 times and the majority-vote score recomputed per resample; the SD of the resampled scores is reported as $\pm$ in the experiment tables — and across every configuration in this paper that SE is at most 0.5 events. What protects the comparisons beyond the SEs is that they are paired (all configurations share seeds 100–107) and that every conclusion below rests on a

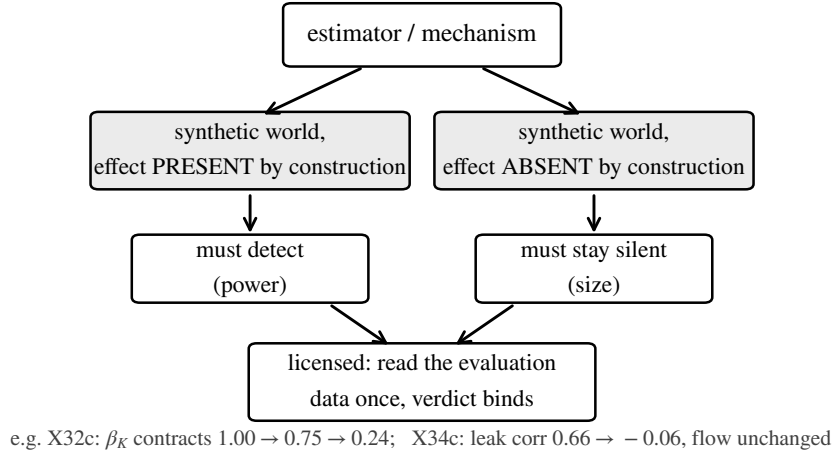


Figure 1. The validation pattern. Every estimator and every tested mechanism must, before any evaluation data is read, both detect a planted effect on a synthetic world where the effect exists (power) and remain silent on a world where it does not (size). The examples are this paper’s: the fixed-point operator’s contraction gate and the quote-skew maker’s absorption gate.

pre-registered refutation criterion, not on a one-event margin.

4.2 Size and power validation on synthetic ground truth

Every estimator in the program passes a validation gate before touching evaluation data: a test on synthetic data where the answer is known, establishing both size (on a world where the effect is absent, the test does not fire) and power (on a world where the effect is present by construction, it does) (Figure 1). The battery’s estimators were licensed this way before the first agent-based run, and — the step specific to this paper — so was every *tested mechanism*: the anticipator was shown to profit and damp flow impact on a synthetic world (Section 5.2), the fixed-point stack to contract forecastable flow (Section 5.3), the quote-skewing maker to absorb impact while leaving the flow untouched (Section 5.4). This is what makes the refutations evidential: when a licensed mechanism fails to move the battery, the failure is a property of the market, not of a broken implementation. At the time of writing the continuous-integration suite runs all 39 verification gates (the numbered gates X1–X35 plus four unnumbered platform gates) on every commit.

4.3 Pre-registration, budgets, and tie-breaks

“Pre-registered” here means: the design document — hypotheses, budgets, refutation criteria — is committed to the repository before any evaluation run, and the version-control history is the timestamp; there is no external registry. Each experiment fixed, before any evaluation run: hypotheses (with one designated primary), the full parameter set, the evaluation protocol (8 seeds, IDs 100–107, $T = 6000$, majority vote per event, identical across all three experiments so ablations are paired seed-for-seed), a refutation (“kill”) criterion, and a bounded tuning budget — one grid pass on disjoint tuning seeds (900–903), permitted only under a pre-declared trigger, with a deterministic pre-declared tie-break (“weakest intervention wins”). Amendments were permitted only before evaluation data was read, and each is recorded in the registration

documents with its trigger. Two features of this protocol proved consequential. First, the refutation criteria fired three times and were honored three times. Second, the tie-break rule turned the tuning passes themselves into evidence: in all three experiments the battery’s preferred setting was the weakest intervention available — the market scores best when the rational agent barely acts — which is the structural claim visible in the selection pattern before it is visible in the scores.

4.4 Reproducibility discipline

Three practices matter for the trustworthiness of small numeric differences. (1) *Byte-identity pins*: every new mechanism ships behind a flag, and a gate asserts that flag-off output is byte-identical to the pre-change simulator, so the calibration (SPY 10/10, GBM 3/10) and all previously published rows are protected across the entire experimental sequence; Experiment II’s $K=1$ reproduces Experiment I’s published configuration exactly, and Experiment III’s $\lambda_Q=0$ reproduces both. (2) *Causality gates*: tampering with future inputs must leave the trade or quote prefix bit-identical (and the tail must differ, proving the mechanism actually reads the tape). (3) *Architecture-portable assertions*: an early version of the pins asserted absolute byte hashes of floating-point output, which held on one CPU architecture and failed on another by units in the last place; the discipline that survives is to assert in-process equivalence and architecture-stable invariants rather than absolute hashes.

Separately from this paper’s claims: performance claims elsewhere in the same research program are governed by a trial ledger and reported as deflated Sharpe ratios (Bailey and López de Prado, 2014) beside a combinatorially cross-validated probability of backtest overfitting (Bailey et al., 2017); the program’s standing rule is that its PBO of 0.45 is restated beside any Sharpe claim. This paper makes mechanism claims, not performance claims, but the same accounting culture (Harvey et al., 2016) — count the trials, publish the failures — produced it.

5 The model

5.1 The base market

The base market is deliberately small: aggregated flows, about a dozen parameters, linear impact. Log price p_t moves by

$$r_t = \lambda [D_t^{\text{fund}} + D_t^{\text{chart}} + D_t^{\text{vol}} + D_t^{\text{mm}} + D_t^{\text{noise}}], \quad (1)$$

with price-impact coefficient λ ; $D_t^{\text{fund}} = k_F(V_t - p_t)$ for a random-walk log fundamental V ; $D_t^{\text{chart}} = k_C \tanh(m_t/s)$ for an EWMA trend signal m with saturation scale s ; $D_t^{\text{vol}} = k_V(L_t - L_{t-1})$ for volatility-targeters (hereafter vol-targeters) running the leverage rule $L = \min(L_{\text{max}}, \sigma^*/\hat{\sigma})$ against an EWMA variance estimate $\hat{\sigma}^2$; a one-lag market-maker flow $D_t^{\text{mm}} = -k_M r_{t-1}$; and i.i.d. noise. The volatility-targeting flow is the reflexive ingredient: a vol spike forces de-leveraging (selling), which moves price, which raises vol — the minimal form of the leverage-constraint feedbacks of Brunnermeier and Pedersen (2009) and Adrian and Shin (2010). The market-maker ingredient was added at the calibration stage, before any ablation was scored, when tuning showed that vol-targeting flows structurally leak momentum — their de-leveraging is serially correlated — and that in reality such flow is absorbed by liquidity provision.

Parameters were set by hand once, on the full configuration, to produce realistic volatility levels and pass E1–E3 only, and were not tuned per configuration or per event; ablations zero out flows and change nothing else. Eight seeds per configuration, $T = 6000$; an event passes if a majority of seeds pass.

5.2 The anticipatory agent (Experiment I)

The vol-targeters’ flow is mechanical and forecastable from public information alone: their mandate ($L = \min(L_{\max}, \sigma^*/\hat{\sigma})$) and their estimator (an EWMA of squared returns at a known speed) are both computable from the price tape. The anticipatory agent A is given exactly that information and nothing else. At each step it reconstructs $\hat{\sigma}_t^2$ from the return history, forecasts the integrated future mechanical flow under the one belief that defines vol-targeting — volatility reverts to the targeters’ own target, so leverage ends at $L_{\text{eq}} = \min(L_{\max}, 1)$ —

$$\hat{F}_t = k_V \cdot \text{mean}_{\text{cohorts}}(L_{\text{eq}} - L_t), \quad (2)$$

holds inventory proportional to the forecast, capped by capital, $I_t^* = \text{clip}(k_A \hat{F}_t, \pm \text{cap}_A)$, and trades the inventory change plus execution noise of scale s_A . The unwind is automatic: as vol reverts and the targeters re-lever, the forecast decays and the anticipator sells into exactly that mechanical bid. Three parameters (k_A, cap_A, s_A) — response, capital, noise — frozen before any battery run.

Three gates license the mechanism before any score is read (the pattern is described in Section 4.2): (a) with the agent disabled the simulator reproduces the original market byte-for-byte, protecting the SPY-10/GBM-3 calibration; (b) the forecast is causal — tampering with future returns leaves the trade prefix unchanged; (c) on a deterministic synthetic world with a perfectly predictable mechanical flow, the anticipator profits *and* damps the flow’s price impact, from 1.00 to 0.75 per unit of arriving flow. The gates establish that the agent functions as designed. The pre-registered refutation criterion: if the frozen configuration does not exceed 5/10 on the evaluation seeds, the expectation hypothesis is refuted.

The parameter protocol matters for what follows. The first-shot parameters (k_A, cap_A, s_A) = (0.5, 0.02, 0.002) scored 4/10 on tuning seeds disjoint from evaluation, so the single pre-registered grid pass was spent: its best score was 5/10, tied across all six $k_A = 0.25$ combinations plus one weak $k_A = 0.5$ setting — flat wherever the anticipator is weak. The pre-declared deterministic tie-break (lexicographically smallest, i.e. the weakest anticipator) selected (0.25, 0.01, 0.001). That the selection lands on “trade as little as possible” was recorded, at registration time, as already being evidence about the hypothesis.

5.3 The fixed point of mutual anticipation (Experiment II)

The fixed-point stack makes each layer’s unwind part of what is anticipated. Layer k front-runs the *residual* forecastable total flow — the mechanical flow plus the deterministic future unwind of the $k - 1$ layers beneath it (time subscripts suppressed):

$$\text{resid}_0 = \hat{F}, \quad J_k = \text{clip}(k_A \text{resid}_{k-1}, \pm \text{cap}_A), \quad \text{resid}_k = \text{resid}_{k-1} - J_k, \quad I_K = \sum_{k=1}^K J_k. \quad (3)$$

Away from the caps this telescopes: $I_K = (1 - (1 - k_A)^K) \hat{F}$, and the model-implied forecastable residual of total flow contracts geometrically, $\text{resid}_K / \hat{F} = (1 - k_A)^K \rightarrow 0$. $K = 0$ is the flow-only

control; $K = 1$ is byte-for-byte Experiment I’s single layer; $K \rightarrow \infty$ is the rational-expectations limit of this operator class. The budget was fixed before any run: $K = 5$ exactly, no convergence-based stopping, evaluation on the same eight seeds, and a contingent six-candidate tuning pass permitted only on a regression below the control’s 5/10, with the same weakest-anticipation tie-break.

One amendment was made at the gate stage, before any battery run, and it illustrates what the gates are for: the first formulation capped the *stack*’s total inventory at a single agent’s cap_A , and the pre-specified mechanism gate failed it — under a shared cap the $K=5$ stack holds exactly the $K=1$ inventory through every large episode and the measured forecastable-flow contraction disappears. A total-cap stack cannot express the hypothesis it exists to test. The licensed formulation caps each layer separately.

With per-layer caps the mechanism gate passes: on the deterministic synthetic world the projection of realized total flow onto the mechanical flow contracts as $\beta_K = 1.000 \rightarrow 0.750 \rightarrow 0.237$ at $K = 0 \rightarrow 1 \rightarrow 5$, matching the $(1 - k_A)^K$ theory line exactly. The operator does what the rational-expectations argument says it should. The question is whether the market obeys the argument.

5.4 The price-setting market maker (Experiment III)

Experiments I and II eliminate every agent that *trades* against the forecastable flow: an anticipator’s inventory is driven by the same slow public state, so its trades re-leak what they absorb. One hypothesis survives that diagnosis: *price-setting* rationality — a market maker who does not trade but quotes, pre-adjusting the price by the expected impact of the forecastable flow so that the information is absorbed into the price *level* rather than left in the return. The maker reuses the same causal public-state forecast $\hat{F}_t$ of Eq. (2), holds no inventory, posts no flow, and draws no noise:

$$q_t = \lambda_Q \lambda \hat{F}_t, \quad r_t = \lambda D_t + (q_t - q_{t-1}), \quad (4)$$

with D_t the total flow in brackets in Eq. (1) and skew strength $\lambda_Q \in \{1.0, 0.5\}$ fixed in advance ($\lambda_Q = 0$ is byte-identical to the unmodified simulator). Because the maker’s forecast state *is* the targeters’ public state, the quote revision telescopes exactly against the mechanical flow $f_t^{\text{mech}} = D_t^{\text{vol}}$, $q_t - q_{t-1} = -\lambda_Q \lambda f_t^{\text{mech}}$: at full skew the mechanical flow’s entire price impact is absorbed into the level by algebraic identity, not by tuning, and the realized return keeps only the unforecastable flow surprise. Nothing is left that could re-leak — no inventory, no unwind, no cap, no execution noise. The flow itself still executes untouched; only where its impact lands (level versus return) changes.

The mechanism gate verifies this to the identity on the deterministic synthetic world: at $\lambda_Q = 1$ the leak estimator $\text{corr}(f_t^{\text{mech}}, r_{t+1})$ collapses from +0.66 to −0.06 while the mechanical-flow series itself is essentially unchanged — absorption into the level, not suppression of the flow. The registration declared this the third and final attempt on the expectation line and included a closure clause: if the primary hypothesis is refuted at this budget, the 5/10 ceiling is declared structural for this model class and the line closes.

	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	
SPY (real)	■	■	■	■	■	■	■	■	■	■	10/10
G	■	□	□	□	□	■	□	□	■	□	3/10
F	■	□	□	□	□	■	□	□	■	□	3/10
FC	□	□	□	□	□	■	□	□	□	□	1/10
FV	□	■	□	□	■	■	■	□	□	■	5/10
FCV	□	■	□	□	■	■	■	□	□	■	5/10
FCVM	■	■	□	□	■	■	□	□	□	■	5/10
FCVMH	■	□	□	□	■	■	□	□	□	■	4/10

Figure 2. Battery scorecard: real SPY against the ablation ladder (filled = pass). G = noise only (GBM); F adds fundamentalists; C chartists; V volatility-targeters; M market makers; H multi-horizon heterogeneity. No flow configuration exceeds 5/10, and the five events missing from the best configuration (FCVM) are E3, E4, E7, E8, E9.

	$AC_1(r)$	kurtosis	$AC_1(r)$	leverage	E9 bits
SPY (real)	-0.103	15.1	0.31	-0.113	0.0029
GBM (G)	0.001	3.0	-0.001	0.007	0.0001
FCVM (best flow-only)	0.036	8.8	0.38	-0.125	0.0184

Table 3. Key medians. FCVM reproduces efficiency, tails, clustering amplitude and the leverage effect at realistic magnitudes; what it cannot do is keep the sign channel silent (E9: 0.0184 significant direction bits against SPY’s 0.0029) or generate long memory and the coupling arrow.

6 Results

6.1 The ablation ladder and the five-event ceiling

Figure 2 shows the ladder. Noise alone (G) and fundamentalists alone (F) score the Gaussian 3/10. Chartists without liquidity provision are destabilizing: FC scores 1/10, because the momentum leak destroys even efficiency. The vol-targeters are the decisive ingredient: FV jumps to 5/10, generating the asymmetric amplitude facts — fat tails (E2), the leverage effect (E5), clock jumps (E7), crash asymmetry (E10) — because the de-leveraging spiral is the only flow that reacts to volatility and reacts in one direction only. Adding market makers (FCVM) restores efficiency (E1) at the cost of E7, holding the score at 5/10: efficiency and the amplitude facts are contributed by *different* agents. Multi-horizon heterogeneity (FCVMH) fails both ways, dropping to 4/10.

The ceiling is 5/10 and the five missing events constitute a diagnosis (Table 3). FCVM fails E3 (slow volatility clustering), E4 (long-memory clock), E7 (clock jumps, marginally), E8 (the arrow in the coupling), and E9 (information-free signs). The E9 failure is central to the remainder of the paper, so we give it a name used throughout: the *sign leak* — a

Config	Score	Failed events	E9 bits (median)
FCVM (control)	5/10 $\pm$ 0.18	E3 E4 E7 E8 E9	0.0184
FCVM+A (hypothesis)	5/10 $\pm$ 0.03	E3 E4 E7 E8 E9	0.0200
FV+A	5/10 $\pm$ 0.10	E1 E3 E4 E8 E9	0.0158
F+A (no targeters)	3/10 $\pm$ 0.16	= F exactly	0.0002

Table 4. Experiment I ablation (8 evaluation seeds, majority vote; $\pm$ is the seed-bootstrap SE of the score, Section 4.1). Adding the anticipator changes none of the targeted quantities: the failure set is identical and the direction bits do not fall.

significant excess of E9’s statistic, the *direction bits* $I(\text{tape features; sign } r_{t+1})$, over its shuffle null. FCVM leaks 0.0184 bits where real SPY shows 0.0029. The leak has a known source: the vol-targeters’ de-/re-leveraging drift is serially correlated and forecastable from the tape, and a one-lag market maker cannot absorb a multi-day drift. The natural hypothesis — the one the three experiments test — is that the missing ingredient is *expectation*: rational agents who anticipate the predictable flow should arbitrage the sign leak away and perhaps stretch the volatility episodes into long memory.

6.2 Experiment I: one anticipator

Table 4 shows the outcome. The refutation criterion is met: FCVM+A scores 5/10 with a failure set identical to the control’s. The events expectation was hypothesized to supply did not move — the direction bits did not move toward zero (medians $0.0184 \rightarrow 0.0200$; the paired comparison, extended to 32 seeds under a registered fixed budget, is statistically flat — the anticipator above control on 19 of 32 seeds, two-sided Wilcoxon $p = 0.57$ — so the licensed reading at one layer is “unchanged,” not “reduced”), and the long-memory events stayed flat. The interaction hypothesis — that the anticipator can act only through the flow it fronts — held: without vol-targeters the anticipator contributes nothing ($F+A = F = 3/10$) — expectation requires a predictable flow to trade against; having one, however, repairs none of the missing events.

The reason the attempt fails is itself the finding. The tuning grid is monotone (Figure 3): the harder the anticipator trades, the worse the market scores. Efficiency breaks (its unwind adds reversal: AC_1 falls from -0.09 at the frozen setting toward -0.35 at $k_A = 1$), the tails erode (it damps the very spiral that generates them, kurtosis $8.8 \rightarrow 4.6$ from the control to the grid’s most capitalized settings), and the sign leak *grows* (up to 0.039 bits at full strength $k_A = 1$). The full grid — per-setting scores and statistics — is exported with the archived results.

The mechanism is structural, not parametric: the anticipator’s inventory is a function of the same slow public volatility state as the flow it front-runs, so its own future unwind is exactly as forecastable as what it absorbs. One layer of expectation does not remove the predictable flow; it reproduces the leak one derivative earlier, while eroding the amplitude facts. The refutation names its own successor: if one anticipator re-leaks because its inventory is public-state-driven, perhaps information-free prices require anticipators who anticipate *each other* — the fixed point of mutual anticipation.

$k_A = 0.25$	5	5	5	5	5	5
$k_A = 0.5$	5	4	4	4*	3	3
$k_A = 1$	4	4	4	4	4	4
	0.01/0.001	0.01/0.002	0.02/0.001	0.02/0.002	0.05/0.001	0.05/0.002
	cap_A / s_A					

Figure 3. The Experiment I tuning grid (tuning seeds 900–903, disjoint from evaluation): battery score by anticipator response k_A , capital cap_A , and execution-noise scale s_A . Darker cells score higher. The score is non-increasing in every strengthening direction; the tied best settings are the weakest anticipators. The heavy box marks the tie-break selection (the frozen evaluation setting); the asterisk marks the pre-registered first-shot setting.

Config	Score	Failed events	E9 bits (median)
$K=0$ (FCVM, control)	5/10 $\pm$ 0.18	E3 E4 E7 E8 E9	0.0184
$K=1$ (Exp. I layer, byte-identical)	5/10 $\pm$ 0.03	E3 E4 E7 E8 E9	0.0200
$K=5$ (frozen carry-over)	3/10 $\pm$ 0.49	E1 E3 E4 E5 E7 E8 E9	0.0257
$K=5$ (tuned, contingent pass)	5/10 $\pm$ 0.00	E3 E4 E7 E8 E9	0.0168

Table 5. Experiment II: the K -ladder. The median sign leak grows in K — the depth-five rise holds on 7 of 8 paired seeds, while the one-layer step ($K=0 \rightarrow 1$) is statistically flat in a registered 32-seed extension (Wilcoxon $p = 0.57$): the licensed reading is the depth trend, not the single-layer increment — and the model-space forecastable flow contracts toward zero; the frozen $K=5$ stack is a two-event regression (it breaks efficiency and removes the leverage effect).

6.3 Experiment II: the inversion

Table 5 and Figure 4 give the result, which is the paper’s central experiment. The refutation criterion is met again, and this time the two diagnostics move in *opposite directions*: as K goes $0 \rightarrow 1 \rightarrow 5$, the model-space forecastable flow falls $1.000 \rightarrow 0.750 \rightarrow 0.237$ while the market-space sign leak rises $0.0184 \rightarrow 0.0200 \rightarrow 0.0257$ bits in the medians, with $K=5$ above the control on 7 of 8 paired seeds. (The one-layer step inside that rise is a point estimate within seed noise — the registered 32-seed extension is statistically flat, medians 0.0195 versus 0.0194, Wilcoxon $p = 0.57$ — so the ladder licenses the depth trend, not the single-layer increment.) The frozen $K=5$ stack is moreover a two-event regression: its aggregate unwind adds reversal (AC_1 : $-0.075 \rightarrow -0.247$, breaking E1) and it absorbs the very spiral that generates the leverage effect (E5 fails, $-0.095 \rightarrow -0.030$), while tails erode (kurtosis $6.8 \rightarrow 5.3$) and crash asymmetry collapses (tail ratio $44.8 \rightarrow 4.3$). Rational front-running at depth does not merely fail to supply the missing events; at carry-over strength it removes amplitude facts the market already had.

The regression triggered the pre-registered contingent pass: six candidates on disjoint tuning seeds returned a best score of 5/10, tied across the three weakest settings, and the pre-declared tie-break again selected the weakest — $k_A = 0.05$, an effective stack strength of

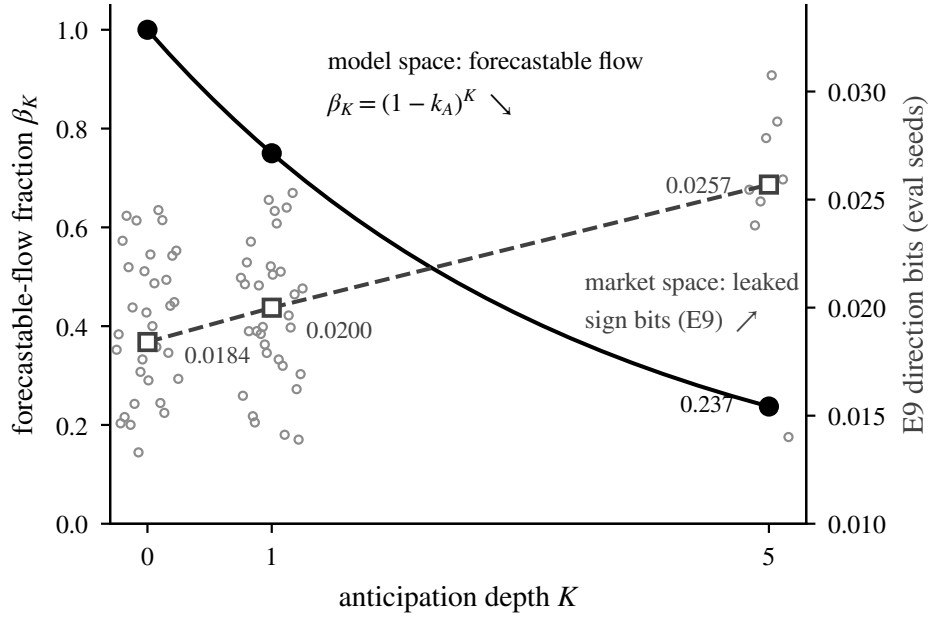


Figure 4. The inversion. Solid curve and filled circles (left axis): the forecastable-flow fraction β_K measured on the deterministic synthetic world, falling exactly as $(1 - k_A)^K$ — $1.000 \rightarrow 0.750 \rightarrow 0.237$. Dashed line and open squares (right axis): the E9 direction bits measured in the simulated market at the same K (open circles: per-seed values — 32 seeds at $K \in \{0, 1\}$ from the registered extension, the 8 evaluation seeds at $K=5$), medians rising $0.0184 \rightarrow 0.0200 \rightarrow 0.0257$; the $K=5$ rise holds on 7 of 8 paired seeds, while the one-layer step is flat at 32 seeds. The operator removes forecastable flow in model space; the market re-creates sign information in price space. Open-loop contraction does not survive closed-loop equilibrium.

$1 - 0.95^5 = 0.226$, weaker than Experiment I’s single frozen layer (0.25). Evaluated once on the evaluation seeds, the selected setting scores 5/10, failing exactly the control’s five events. The selection pattern of Experiment I repeats one level up: the tuning criterion rewards weaker anticipation at every opportunity, never stronger.

The algebra explains the failure sharply. Away from the caps, the K -layer stack is equivalent to *one* anticipator of strength $1 - (1 - k_A)^K$ (the telescoped sum in Eq. (3)): mutual anticipation of this operator class deepens into a stronger single anticipator, and Experiment I already measured that stronger is worse. The contraction that defines the fixed point is an open-loop property, licensed by the gate against an exogenous tape; it does not survive closed-loop equilibrium, because every layer’s inventory is a function of the same slow public volatility state, so the stack’s own price impact regenerates exactly the forecastable structure the model says it removes. Model-space forecastable flow down, market-space sign leak up: this is, to our knowledge, the sharpest available statement of why the rational-expectations limit cannot be reached by stacking boundedly-capitalized flow traders.

6.4 Experiment III: absorption removes the amplitude facts, not the leak

Table 6 and Figure 5 give the sharpest refutation of the three. The full-skew maker removes the vol-targeters’ forecastable impact exactly — and the sign leak does not close. Bits stay flat at half skew (0.0180 versus 0.0184) and *grow* at full skew (0.0237, higher than the control on 7 of

Config	Score	Failed events	E9 bits (median)
FCVM (control)	$5/10 \pm 0.18$	E3 E4 E7 E8 E9	0.0184
FCVM+Q($\lambda_Q=1.0$) (theory case)	$1/10 \pm 0.34$	all but E6	0.0237
FCVM+Q($\lambda_Q=0.5$)	$3/10 \pm 0.03$	E1 E2 E3 E4 E7 E8 E9	0.0180
FCVM+Q($\lambda_Q=0.05$) (tuned, contingent)	$5/10 \pm 0.00$	E3 E4 E7 E8 E9	0.0176

Table 6. Experiment III. Full skew removes the vol-targeters’ forecastable impact exactly, and the sign leak does not close — it grows (0.0237, higher than the control on 7 of 8 seeds) — while the amplitude facts disappear.

8 seeds). The leak was never the vol-targeters’ drift alone: Experiment I’s original attribution is overturned. It lives distributed across *all* the state-dependent flows — chartist momentum, fundamentalist reversion, the one-lag maker’s partial absorption — and deleting the spiral’s impact strips away the amplitude structure that was masking them: efficiency breaks ($AC_1: +0.04 \rightarrow -0.26$; with the spiral gone the liquidity provider’s reversion dominates), and the newly-Gaussian returns make the remaining structure easier to detect.

The second pre-registered hypothesis — that the maker would neutralize only the forecastable *mean* of the flow, leaving the volatility dynamics intact — is also refuted at the theory case, and its refutation is the paper’s central fact: in this market the forecastable flow *is* the generator of the amplitude facts. Full absorption removes the fat tails (kurtosis $8.8 \rightarrow 3.0$), the leverage effect ($-0.125 \rightarrow -0.008$), crash asymmetry (tail ratio $40 \rightarrow 1.1$) and volatility clustering ($AC_1(|r|)$ $0.38 \rightarrow 0.06$) simultaneously. Half skew regresses as well ($3/10$), so the contingent pass fired: the six-candidate grid returned a uniform $5/10$ for every $\lambda_Q \leq 0.30$, and the pre-declared weakest-skew tie-break picked $\lambda_Q = 0.05$ — the same selection pattern a third time, now for a pricing rule. Evaluated once on the evaluation seeds, the selected setting scores $5/10$, failing exactly the control’s five events. Across all three experiments and all three agent classes, the tuning criterion selects the weakest rational intervention available.

Figure 6 collects the E9 evidence across all three experiments.

6.5 Composition: the structural claim

The three refutations compose. In this model class, the forecastable flow component and the asymmetric amplitude facts are the same object:

1. an agent who *trades* against the forecastable flow adds its own forecastable flow, because its inventory is driven by the same slow public state — the leak moves one derivative earlier (Experiment I);
2. stacking such agents to the fixed point of mutual anticipation is algebraically one stronger such agent, and re-leaks harder while the model-space forecastable flow contracts to 0.237 (Experiment II);
3. a price-setter who absorbs the flow’s impact exactly — the rational-expectations limit for this channel, verified to the identity — can close the leak only by deleting the de-leveraging dynamics that generate the tails, the leverage effect, and the crash asymmetry, scoring $1/10$ (Experiment III).

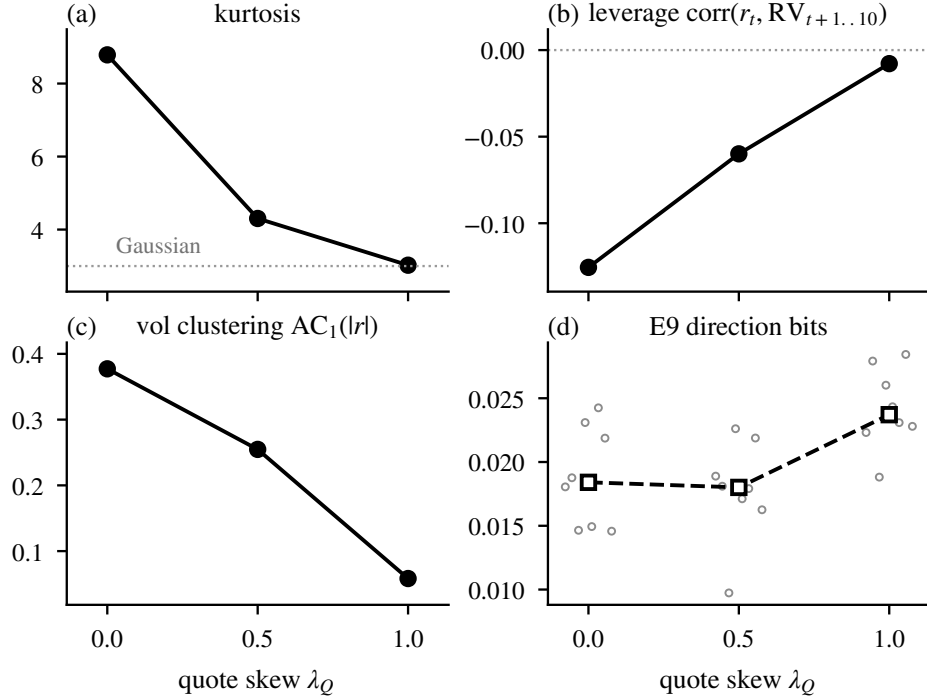


Figure 5. Amplitude-fact medians versus quote skew λ_Q (8 evaluation seeds). Full absorption of the forecastable flow’s price impact removes the fat tails (kurtosis $8.8 \rightarrow 3.0$), the leverage effect ($-0.125 \rightarrow -0.008$) and volatility clustering ($AC_1(|r|)$ $0.38 \rightarrow 0.06$) — while the E9 direction bits (panel d; open circles are seeds) do not close: flat at half skew, higher at full skew.

Sign efficiency and the amplitude facts are not separate dials: in a market moved by state-dependent flows they are jointly produced, and no added layer of rationality — trading, iterated, or price-setting — supplies the five missing events (E3, E4, E7, E8, E9). What the minimal market lacks is not a smarter agent but a different kind of memory: the long-memory clock and the coupling arrow live upstream of the flows, in whatever generates them.

7 External validity: the leverage effect across market structures

The model mechanism makes the amplitude channel a property of the *flow structure*: the leverage effect in the minimal market exists because a one-directional, state-dependent deleveraging flow exists, and disappears when that flow’s impact is removed (Figure 5). A weak but testable empirical counterpart follows: across real asset classes, the leverage effect should follow the microstructure that does or does not host such flows.

Table 7 and Figure 7 give the measurement, using an estimator battery identical across the three vertices (realized variance from daily OHLC ranges via Garman–Klass (Garman and Klass, 1980)). The classical readings of the equity effect — firm capital structure (Christie, 1982) and volatility feedback through the equity risk premium (Bekaert and Wu, 2000) — are equity-specific mechanisms, which is exactly what makes the cross-venue comparison informative: a currency is not a levered claim and carries no equity premium, so whatever asymmetry survives outside equities must come from the venue’s flow structure. Equity markets — where financial

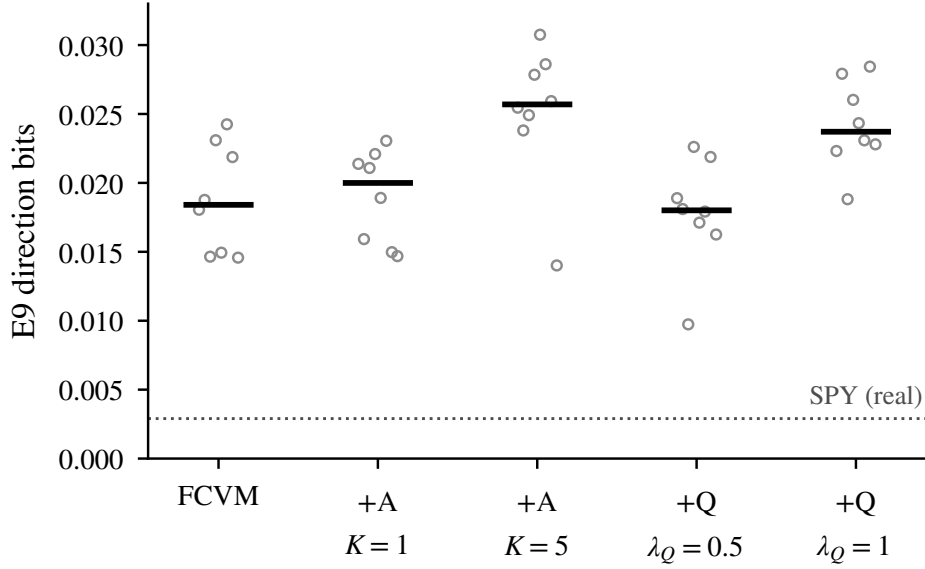


Figure 6. E9 direction bits across every rationality configuration (open circles: evaluation seeds; bars: medians). No configuration approaches the information-free regime; real SPY (dotted, 0.0029) sits far below all of them. Trading against the flow, iterating that trade to depth five, and absorbing the flow’s impact into the price level all leave the sign channel open or wider.

	equity cohort	FX	crypto
leverage effect	-0.0405 ± 0.0079	$+0.005 \pm 0.009$	$+0.031 \pm 0.016$
z vs. zero	-5.1	0.56	1.97
z vs. FX	3.89	—	1.44
instruments positive	0 of 4 markets	7 of 13 pairs	8 of 10 coins

Table 7. The leverage triangle. Equity cohort: four universes (US, Japan, Europe, Asia-EM), 2010–2026. FX: 13 liquid crosses, 2010–2026. Crypto: 10 major coins, 2017–2026. Each z vs. zero is the cohort mean divided by its cross-sectional dispersion (the $\pm$ values in the first row). The ordering $-0.041 < +0.005 < +0.031$ is monotone in point estimates; the equity edge is statistically certified, the FX–crypto edge is inside one-sided noise.

leverage and institutional de-risking cascades live — show the textbook negative effect in all four universes (-0.030 to -0.047). FX — an institutional dealer market hosting neither de-risking cascades nor a dominant retail-momentum dynamic — is statistically zero ($+0.005$, z vs. zero = 0.56; 7 of 13 pairs positive, the symmetric scatter a true zero predicts). Crypto — no financial leverage, no institutional de-risking, a retail dynamic in which rallies rather than crashes spike volatility — sits at $+0.031$, with 8 of 10 coins individually positive and only BTC (-0.043) and ETH (-0.019) retaining the equity-style sign.

A distinction between what is statistically certified and what is suggested by point estimates is required here: the equity edge is resolved (one-sided $z = 3.89$ against FX, $z = 4.06$ against crypto), and FX is decisively zero; but the FX–crypto edge is $z = 1.44$, inside noise, and crypto itself clears zero at only $z \approx 2$ — so “crypto genuinely positive above an FX zero” is an ordering of point estimates, not a certified separation.

That edge is not under-analyzed; it is underpowered, and the bound is arithmetic. Certifying

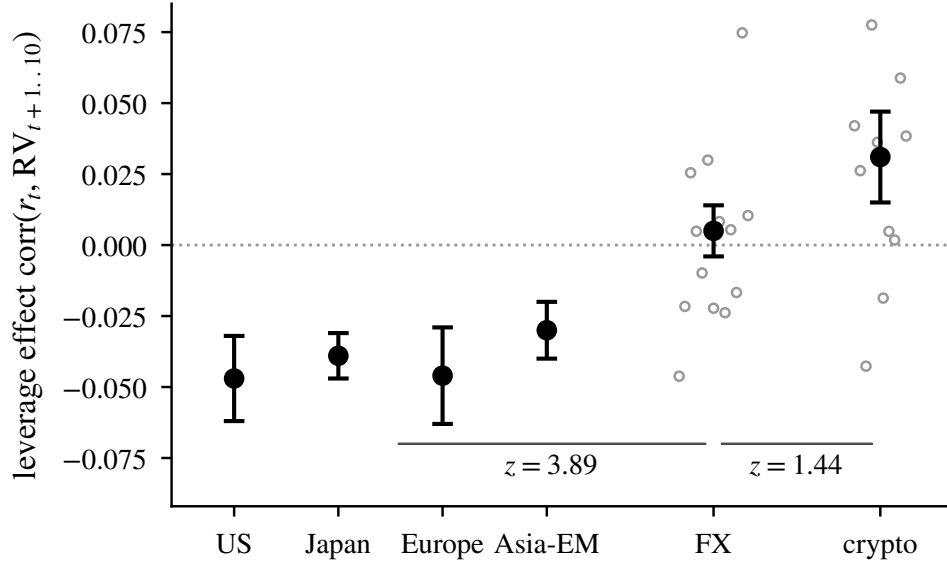


Figure 7. The leverage effect $\text{corr}(r_t, \text{RV}_{t+1..10})$ at three microstructure vertices. Filled points: cohort estimates ± 1 s.d. Open circles: the 13 individual FX crosses and 10 individual coins. Equities (four universes: US, Japan, Europe, Asia-EM) sit at -0.030 to -0.047 ; FX at $+0.005 \pm 0.009$ (statistically zero, $z = 0.56$); crypto at $+0.031 \pm 0.016$. The equity–FX separation is resolved ($z = 3.89$); the FX–crypto separation is not ($z = 1.44$).

$z \geq 2$ at the current point separation requires the crypto sampling SD to fall from 0.0158 below 0.0099, a $1.6\times$ reduction; the SDs are time-block bootstrap SDs scaling as $1/\sqrt{T}$, so history alone would need ≈ 22 years of daily crypto data against the 8.6 that exist (major-coin daily history begins in late 2017). Nor can the cross-section substitute for T : a registered widening of the universe from 10 to 17 majors — every liquid 2017-era coin passing the pre-declared real-range and history gates — left the sampling SD essentially unchanged ($0.0158 \rightarrow 0.0161$) while the pooled estimate moved to $+0.026$ (14 of 17 coins positive) and the z against FX to 1.16; coins share their volatility events, so widening adds instruments but almost no independent information. (The equity–crypto separation, by contrast, survives the widening at $z = 3.71$.) At these point estimates the FX–crypto edge becomes certifiable only by the accrual of history — not by more coins, more crosses, or a different test — and we report it as what it is: an ordering the data cannot yet promote to a separation.

The residual per-pair scatter in FX (Figure 8) lies on the flight-to-quality axis: the three most negative crosses are all yen pairs (AUDJPY -0.046 , GBPJPY -0.024 , EURJPY -0.022 — yen strength is risk-off, so “down $\rightarrow$ vol up” is flight-to-quality mimicking the leverage effect), and the most positive is USDMXN ($+0.075$, the same flow with the opposite quote orientation).

The reading is licensed by a sign-recovery gate: on synthetic worlds with a known leverage sign (equity-negative, inverted-positive, symmetric-zero), the estimator recovers each with wide separation and its spurious-leverage bound is $|0.04|$, so a zero is read as a zero and an inversion as an inversion.

Finally, the triangle’s epistemic status is a stated boundary of the paper’s claims, and it is worth being precise about what would move it. The triangle is *consistent with* the thesis — the amplitude channel follows the flow structure of the venue, negative exactly where forced de-risking lives, zero in the institutional market without it, and (in point estimate) positive

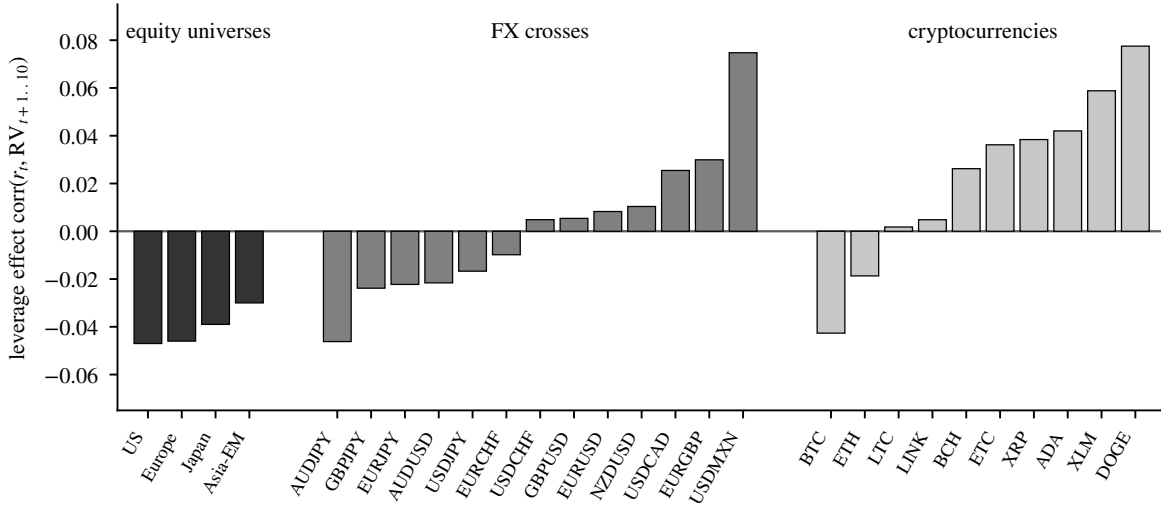


Figure 8. The leverage effect per instrument, grouped by venue class: the four equity-universe estimates, the 13 FX crosses, and the 10 registered cryptocurrencies, each sorted within its group. Every equity universe is negative; the FX crosses scatter symmetrically about zero along the flight-to-quality axis; the coins are predominantly positive, with BTC and ETH the exceptions.

where an opposite retail dynamic dominates — but consistency with three ordered points is weak evidence by construction: any monotone story over the same venue ordering fits them. What would count as proof is a *discriminating* cross-venue prediction, and the mechanism makes one: a venue whose flow structure changes should see its leverage sign migrate. Crypto’s institutionalization is the live case — the mechanism predicts its leverage effect drifts toward the equity sign as leveraged institutional flow displaces retail momentum, and BTC (−0.043) and ETH (−0.019), the two most institutionally held coins, already carry the equity-style sign in the registered universe (joined by XMR in the widened one) — an in-sample observation that is consistent but proves nothing. Delivering that prediction as a certified test needs either the history the power bound above shows does not yet exist, or flow decompositions (who is trading, not just what the price did) that public daily data does not carry. So the claim stays scoped: the triangle is an empirical counterpart, not an empirical test, and no claim is made that the minimal market’s specific mechanism generates the real effect.

8 Discussion and limitations

8.1 Scope of the claims

The structural claim of Section 6.5 is scoped, and the scoping is part of the result. The following are explicitly *not* claimed: (i) anything about real markets’ generating process; (ii) anything beyond this model class — no learning agents, no heterogeneous information, no strategic equilibrium or rational-expectations solution concept; (iii) optimality of the agent designs — each is gate-verified to *function*, not to be optimal.

“Structural” here means: invariant over the searched class under the pre-registered budgets, with each searched member licensed by a mechanism gate proving it expresses its hypothesis. The claim would be falsified by an agent class outside the three — learning agents that condition

on more than the public volatility state, or information-heterogeneous agents whose inventories do not ride a common public state — breaking the ceiling under the same battery and the same protocol. That is an open research direction, and we name it as such rather than hedge the claim through it.

8.2 The closure clause as a stopping rule

Experiment III’s registration contained, alongside its hypotheses and budget, a closure clause: this is the third and final attempt on the expectation line; if the primary hypothesis is refuted at this budget, the 5/10 ceiling is declared structural for flow-based minimal markets with boundedly-rational agents of every class here considered, and the line closes — no fourth experiment. The clause is not rhetorical; it is the instrument that converts three negative results into one positive claim. An open-ended sequence of failed attempts supports only “we have not yet found the right agent.” A pre-committed final attempt, refuted, supports a stronger statement about the class that was searched — precisely because the search was budgeted, the budget was spent, and the stopping rule was declared before the last result was known.

8.3 Limitations

Three limitations are measured rather than merely acknowledged. First, the battery’s absolute anchor is a US-equity statement: the registered multi-index audit (Appendix B) shows the joint 10/10 calibration extends across the large-cap US complex but not to non-US single indices, which score 5–7/10 with the clock-structure and sign events recurring among the misses; every conclusion in this paper is comparative — same events, same seeds — and is unaffected, but absolute realism claims for non-US venues are not licensed. Second, the battery scores rest on eight evaluation seeds; the seed-bootstrap standard errors reported in every experiment table bound the resulting score uncertainty at half an event, and the three refutations additionally rest on pre-registered criteria and paired seeds rather than on score margins. Third, the FX–crypto edge of the leverage triangle is underpowered, and Section 7 derives the bound: certification requires a history that will not exist for roughly another decade, and widening the cross-section demonstrably does not substitute. Each limitation is stated with the measurement that establishes it, which is the form of limitation a pre-registered program can offer.

9 Conclusion

A minimal flow-based market reproduces half of the stylized-fact battery — efficiency, fat tails, volatility clustering amplitude, the leverage effect, crash asymmetry — and no rationality mechanism of three pre-registered kinds moves it past 5/10. The refutations identify the reason: the forecastable component of flow that keeps the sign channel open is the same object that generates the heavy-tailed, asymmetric amplitude structure, so removing one removes the other. In this model class, sign efficiency and the amplitude facts are jointly produced. The missing events — long memory, the coupling arrow, information-free signs — appear to live upstream of the flows, in whatever generates them, and breaking the ceiling within the protocol would require an agent class outside the three tested here. The cross-venue leverage evidence is consistent with the mechanism’s central implication — the amplitude channel follows the

venue’s flow structure — while remaining, as measured, an ordering rather than a certified separation.

Code and data availability

Every number in this paper is regenerable from the public repository. The research JSONs consumed by the figures and tables — under `research/`: `decathlon.json`, `decathlon2.json`, `decathlon3.json`, `decathlon4.json`, `fx.json`, `crypto.json`, `crypto_wide.json`, `battery_audit.json`, and `score_se.json` — are versioned in the repository and rebuilt by `run_research.py`; the battery code is identical for real data and simulated output; the gate suite runs in continuous integration on every commit, including the byte-identity pins that protect every published row in this paper; and every figure is generated from the research JSONs by `docs/paper/make_figures.py` with no hand-entered values. The pre-registration documents, including the amendments and the closure clause quoted here, are versioned in `docs/design/`.

References

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#	Statistic	Pass region
E1	$AC_1(r)$	$[-0.15, 0.05]$
E2	$kurtosis(r)$	$[4.5, 40]$
E3	$AC_1(r)$ and $\text{mean}_{k=5..20} AC_k(r)$	≥ 0.12 and ≥ 0.05
E4	$AC_8(w)$ (weekly clock level)	≥ 0.12
E5	$\text{mean}_{\tau=1..10} \text{corr}(r_t, RV_{t+\tau})$	≤ -0.03
E6	$kurtosis(z)$	≤ 5
E7	$\text{skew}(u)$ (weekly clock AR(1) innovations)	≥ -0.35
E8	$EP(r), EP(z)$ (entropy production, bits)	$EP(r)$ significant and $EP(z) \leq 0.75 EP(r)$
E9	direction bits $I(\text{features}; \text{sign } r_{t+1})$	$\leq 95\text{th percentile of 120 shuffles}$
E10	$\#\{r < -2.5s\} / \#\{r > +2.5s\}, s = \text{sd}(r)$	≥ 1.25

Table 8. Exact pass criteria, as implemented. E7’s floor is calibrated against the $\log\text{-}\chi^2$ noise skew of the weekly clock (GBM innovations skew ≈ -0.65 ; real SPY -0.22). E8’s entropy production symbolizes the series into 3 bins, measures 3-gram irreversibility, and tests against 120 block-reversal surrogates (random block offsets, blocks reversed with probability $1/2$). E9’s features are the Cartesian product of three tape observables — $\text{sign}(r_t)$, the sign of 21-day momentum, and the volatility tercile — scored by discrete plug-in mutual information with Miller–Madow bias correction (see Paninski, 2003) against a 120-permutation null; the program’s continuous information channel uses the KSG estimator (Kraskov et al., 2004), but E9’s features are discrete by construction. The volatility tercile is cut on full-sample quantiles: E9 is a measurement of information content, computed identically on real and simulated series, not a trading rule.

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A The battery: exact criteria and calibration statistics

Table 8 states each event’s statistic and pass region exactly as implemented in the repository (`kronos/decathlon.py`, function `battery`; the thresholds were fixed in the DESIGN8 registration and its calibration-phase amendments), and Table 9 gives the full statistic vector at both calibration ends and at the flow-only ceiling. Notation: r_t are daily returns; RV_t is the 5-day rolling mean of r^2 (the close-only volatility clock), $\sigma_t = \sqrt{RV_t}$ (written σ_{5d} in Table 2), and $z_t = r_t/\sigma_t$ the deformed (one-clock) return. The weekly clock is built from *non-overlapping* 5-day means w_j of $\log r^2$ — raw weekly differences carry an MA(1) artifact on which even GBM’s $|\Delta w|$ autocorrelates, so E4 uses the level and E7 uses AR(1) innovations $u_j = w_{j+1} - \bar{w}(1 - \phi) - \phi w_j$ with ϕ the fitted AR(1) coefficient. $AC_k(\cdot)$ is the lag- k autocorrelation. A configuration passes an event if a majority of its 8 evaluation seeds pass.

Two calibration facts anchor the thresholds. GBM’s weekly-clock innovation skew (-0.651) is the $\log\text{-}\chi^2$ noise floor E7 is calibrated against; real volatility’s up-jumps cancel most of it (SPY -0.221). And the E4 level-autocorrelation criterion is what a GJR-GARCH process fails (clock AC_8 median 0.03, maximum 0.09, over the battery’s eight evaluation seeds of the gate suite’s GJR world — every seed below the 0.12 bar): exponential memory decays too fast at eight weeks, which is why E4 is the battery’s long-memory event rather than any Hurst point estimate (the close-only Hurst proxy is biased and is recorded as informational only).

Statistic	Event	SPY (real)	GBM (G)	FCVM
$AC_1(r)$	E1	-0.103	0.001	0.036
$kurtosis(r)$	E2	15.09	3.04	8.79
$AC_1(r)$	E3	0.307	-0.001	0.377
$\text{mean } AC_{5..20}(r)$	E3	0.218	0.002	-0.050
$AC_8(w)$	E4	0.186	-0.018	-0.003
leverage	E5	-0.113	0.007	-0.125
$kurtosis(z)$	E6	2.33	2.13	2.20
$skew(u)$	E7	-0.221	-0.651	-0.408
EP(r) bits	E8	0.0211	0.0002	0.0929
EP(z) bits	E8	0.0141	0.0006	0.1392
direction bits	E9	0.0029	0.0001	0.0184
tail asymmetry	E10	1.56	0.91	39.96

Table 9. The full statistic vector at the battery’s calibration ends (SPY 10/10, GBM 3/10) and at the flow-only ceiling (FCVM, 5/10); GBM and FCVM entries are medians over the 8 evaluation seeds. The table shows *how* each score decomposes: GBM passes only E1, E6, E9; FCVM matches SPY’s magnitudes on E1/E2/E5/E10 but has no clock memory ($AC_8(w) \approx 0$ against SPY’s 0.186), fails E8 on the ratio criterion ($EP(z) > 0.75 EP(r)$: deformation does not reduce its entropy production), and leaks $6\times$ SPY’s direction bits. Its E7 skew (-0.408) sits marginally below the -0.35 floor. SPY’s $EP(z)$ is significant but 33% below $EP(r)$, meeting the ratio criterion.

B The multi-index calibration audit

The joint calibration anchor of Section 4.1 is one index. A registered audit (fixed list, declared before any score was read) ran the identical close-only battery on six further real indices: QQQ, DIA, IWM (2010–2026), and the three locally-listed index ETFs used by the research program’s transfer study — 1306.T (TOPIX), EXW1.DE (EURO STOXX 50), 2800.HK (Hang Seng tracker). The registered prediction — every index $\geq 9/10$ — *failed*, and the failure is itself a measurement (Table 10): the anchor extends across the large-cap US complex (DIA joins SPY at 10/10; QQQ and IWM drop single events), but non-US single indices score 5–7/10, missing the clock-structure and sign events (E7, E8, E9 recur) — consistent with the transfer study’s finding that mechanism transfers across venues while calibration values are local. A hygiene probe recorded in the registration rules out the union-calendar padding of the foreign panels as the cause (native-calendar re-runs score the same or lower). The battery’s absolute anchor is therefore a US-equity statement; all conclusions in this paper are comparative and unaffected. Figure 9 shows the full event-by-index pass/fail matrix.

C Robustness material

Score standard errors. The seed-bootstrap standard-error method of Section 4.1 was applied to every configuration published in this paper (fifteen in total, spanning the ablation ladder and all three experiments). The bootstrap reproduces every published majority-vote score exactly before resampling, and the resulting SEs range from 0.00 to 0.49 events (Figure 10); configurations whose per-seed event counts sit at a majority knife-edge (F, the frozen $K=5$ stack, FCVMH) carry the largest SEs, while the headline comparisons (FCVM, FCVM+A, the tuned configurations) are tight.

	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	
SPY (anchor)	■	■	■	■	■	■	■	■	■	■	10/10
QQQ	■	■	■	■	■	■	□	□	■	■	8/10
DIA	■	■	■	■	■	■	■	■	■	■	10/10
IWM	■	■	■	■	■	■	■	□	■	□	8/10
1306.T	■	■	■	□	■	■	□	□	□	■	6/10
EXW1.DE	■	■	■	■	■	■	□	□	□	■	7/10
2800.HK	■	■	□	□	■	■	□	■	□	□	5/10

Figure 9. The multi-index calibration audit as an event-by-index matrix (filled = pass). The clock-structure and sign events (E7, E8, E9) recur among the non-US misses.

Index	Venue	Score	Failed events
SPY (anchor)	US	10/10	—
DIA	US	10/10	—
QQQ	US	8/10	E7 E8
IWM	US	8/10	E8 E10
1306.T	Japan	6/10	E4 E7 E8 E9
EXW1.DE	Europe	7/10	E7 E8 E9
2800.HK	Hong Kong	5/10	E3 E4 E7 E9 E10

Table 10. The multi-index calibration audit (registered list, no post-hoc swaps; misses reported by name). US large-cap indices reproduce the anchor; non-US single indices do not — the battery’s thresholds are US-calibrated, so its absolute scores certify realism only against US equity data, which is how the paper uses them.

The widened cryptocurrency universe. The registered widening of the crypto panel from 10 to 17 majors (Section 3.3) supports the power analysis of Section 7: it leaves the cross-sectional sampling SD essentially unchanged while moving the pooled leverage estimate to +0.026 (14 of 17 coins positive), which is the measured demonstration that the FX–crypto separation is history-bound rather than cross-section-bound.

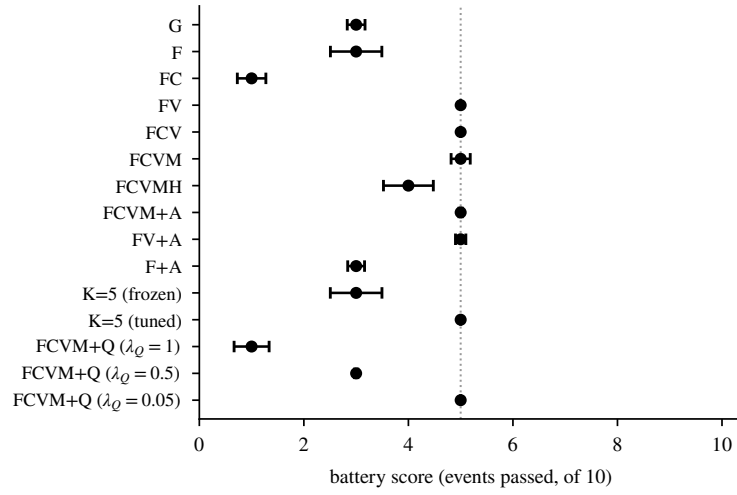


Figure 10. Battery scores with seed-bootstrap standard errors for every configuration published in this paper. The dotted guide marks the 5/10 flow-only ceiling. No SE exceeds half an event.